

1. Questions

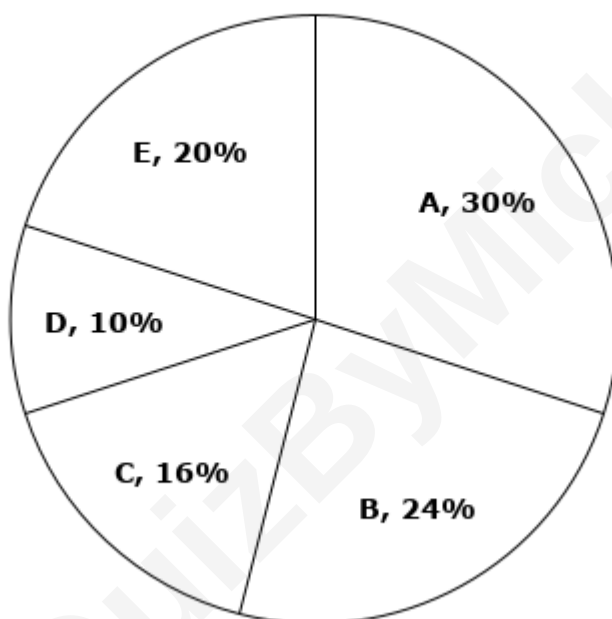
Study the following information carefully and answer the following questions.

The given pie chart shows the percentage distribution of the total number of patients admitted to five different hospitals namely A, B, C, D and E.

Note:

- i). The total number of patients admitted in all hospitals together is 6000.
- ii). The total number of patients admitted in each hospital = Number of male patients admitted + Number of female patients admitted.

Percentage distribution of the total number of patients admitted



The given table chart shows the number of male patients admitted in these five hospitals.

Hospital	The number of male patients admitted
A	900
B	740
C	500
D	200
E	550

In hospital A, out of the total number of patients admitted to the hospital 20% are in critical

condition. Find the number of patients admitted in critical condition.

- a. 420
- b. 300
- c. 360
- d. 280
- e. 200

2. Questions

Find the ratio between the total number of patients admitted in hospital E to the number of female patients admitted in hospital D.

- a. 3:7
- b. 2:1
- c. 5:4
- d. 3:1
- e. 5:3

3. Questions

The total number of patients admitted in hospital F is $(m + 20)\%$ more than that in hospital B. If the total number of patients admitted in hospital A is 72 less than that in hospital F, then find the value of $2m$.

- a. 10
- b. 8
- c. 12
- d. 15
- e. 20

4. Questions

In hospital C, the ratio of the number of male patients admitted for cancer to TB is 3:2. Find the number of male patients admitted for cancer. (Only cancer and TB patients are admitted)

- a. 250
- b. 200
- c. 180
- d. 120
- e. 300

5. Questions

Find the sum of the number of female patients who are admitted in hospital A, B and D together.

- a. 1800
- b. 1600
- c. 2100
- d. 2000
- e. 1500

6. Questions

Study the following information carefully and answer the following questions.

The given table chart shows the average number of students who prefer boating and game zone in four different classes, namely P, Q, R and S, along with the number of students who prefer boating in these four classes.

Class	Average number of students who prefer boating and game zone	The number of students who prefer boating
P	350	250
Q	400	300
R	500	350
S	200	100

In class Q, the ratio of the number of male students who prefer boating to game zone is 2:1 and the number of female students who prefer boating to game zone is 1:4. Find the number of female students who prefer game zone.

- a. 270
- b. 100
- c. 120
- d. 400
- e. 230

7. Questions

The number of students who prefer boating in class Q is what percentage of the number of students who prefer the game zone in class S?

- a. 125%
- b. 200%
- c. 100%
- d. 80%
- e. 70%

8. Questions

The number of students who prefer boating in class R is 50 less than that in class T and the ratio of the number of students who prefer game zone in class T to Q is 3:5. Find the total number of students who prefer boating and game zone in class T.

- a. 650
- b. 700
- c. 600
- d. 1100
- e. 800

9. Questions

Find the difference between the total number of students who prefer boating and game zone in class P and the number of students who prefer game zone in class R.

- a. 200
- b. 80
- c. 70
- d. 50
- e. 100

10. Questions

If the number of students who prefer game zone in class P is $(5m + 50)$, then find the value of m .

- a. 80
- b. 100
- c. 120
- d. 150
- e. 50

11. Questions

A train crosses a woman running in the opposite direction at a speed of 5 m/s in 3 seconds and the same train crosses a car running in the same direction at a speed of 15 m/s in 5 seconds. Find the speed of the train.

- a. 45 m/s
- b. 25 m/s
- c. 20 m/s
- d. 50 m/s
- e. 55 m/s

12. Questions

The total number of students in school A is $x\%$ more than that in school B. If the total number of students in school A is 780, and the ratio of the number of students in school A to B is 13: 10, then find the value of x .

- a. 45
- b. 20
- c. 25
- d. 30
- e. 32

13. Questions

The average age of Mano, Madhan and Mohan is 21 years and the ratio of the age of Mano to Mohan is 3:2. If the age of Madhan is 7 years more than that of Mano, then find the present age of Mohan.

- a. 28 years
- b. 14 years
- c. 21 years
- d. 24 years
- e. 32 years

14. Questions

Rs. 8800 is invested in compound interest at the rate of $(R + 5)\%$ per annum for 2 years. If Rs. 4000 is invested in simple interest at the rate of $R\%$ per annum for 2 years, then the simple interest is Rs. 1200. Find the compound interest received.

- a. Rs. 3872
- b. Rs. 4560

- c. Rs. 2880
- d. Rs. 2899
- e. Rs. 4350

15. Questions

The total cost price of pants and shirt is Rs. 4000. If the pants are sold for a profit of 20% and the shirt is sold for a loss of 10%, then the overall selling price is Rs. 4350, find the cost price of the shirts.

- a. Rs. 1500
- b. Rs. 2000
- c. Rs. 3800
- d. Rs. 3500
- e. Rs. 500

16. Questions

Pipe A and B can fill 20% of the tank in 10 hours and 5 hours respectively. If the efficiency of pipe C is twice that of B, then find the time taken by pipe A and C to fill the tank alternatively.

- a. 18 hours
- b. 10 hours
- c. 12 hours
- d. 20 hours
- e. 5 hours

17. Questions

Amit and Balu started a business by investing in the ratio 5:4. After 5 months, Amit increased his investment by Rs. 3000. At the end of the year, the ratio of the profit share of Amit to Balu is 55:

16. Find the initial investment of Amit.

- a. Rs. 800
- b. Rs. 600
- c. Rs. 1000
- d. Rs. 1200
- e. Rs. 1800

18. Questions

The area of the circle is 616 cm^2 and the length of the rectangle is 6 cm more than the radius of the

circle. If the perimeter of the rectangle is 74 cm, then find the breadth of the rectangle.

- a. 18 cm
- b. 20 cm
- c. 17 cm
- d. 21 cm
- e. 10 cm

19. Questions

The ratio of two numbers A to C is 7:3 and number B is 240% more than the number of C. If the sum of two numbers A and C is 40 less than that of B, then find the value of C.

- a. 800
- b. 500
- c. 1200
- d. 600
- e. 400

20. Questions

Mixtures A and B contain milk and water in the ratio 5:4 and 3:4 respectively. If the quantity of milk in mixture A is 100 litres more than that of mixture B and the quantity of water in mixture A is 200 litres, then find the total quantity of mixture B.

- a. 350 litres
- b. 200 litres
- c. 450 litres
- d. 400 litres
- e. 250 litres

21. Questions

What value should come in the place of (?) in the following questions.

$$25\% \text{ of } 720 + \sqrt{625} - 40\% \text{ of } 1200 = ?$$

- a. -275
- b. -325
- c. -290
- d. -445
- e. -400

22. Questions

$$(\sqrt{576} + 276) / 5 + 30\% \text{ of } 800 = ? * 2$$

- a. 120
- b. 80
- c. 200
- d. 150
- e. 50

23. Questions

$$2400 * 5/16 + 54 * 3 = ?^2 + 12$$

- a. 32
- b. 28
- c. 25
- d. 30
- e. 90

24. Questions

$$60\% \text{ of } 200 - 10\% \text{ of } 500 + 35 = ? * 5$$

- a. 21
- b. 18
- c. 19
- d. 24
- e. 12

25. Questions

$$156 + 244 - 18 * 7 + ? = 80 * 5$$

- a. 215
- b. 135
- c. 126
- d. 88
- e. 72

26. Questions

What approximate value should come in the place of (?) in the following questions.

$$(1332)^{1/3} * 3.98 + ? = 32.12\% \text{ of } 299.99$$

- a. 50
- b. 38
- c. 55
- d. 52
- e. 43

27. Questions

$$[36.21 * 5/6] + 44.89 + 21.2 * 1.97 = ?$$

- a. 107
- b. 88
- c. 117
- d. 45
- e. 90

28. Questions

$$29.99\% \text{ of } 400.19 + 5.12^2 = ?^3 - 71.3$$

- a. 14
- b. 5
- c. 6
- d. 11
- e. 2

29. Questions

$$^3\sqrt{64.23 * 12.13 + 179.98} - ? = 14.12 * 9.89$$

- a. 72
- b. 85
- c. 90
- d. 101
- e. 88

30. Questions

$15.12 * (25)^{1/2} - 18.01\%$ of $200.12 = ?$

- a. 42
- b. 39
- c. 35
- d. 29
- e. 19

31. Questions

What value should come in the place of (?) in the following number series?

?, 56, 28, 196, 98, 686

- a. 7
- b. 12
- c. 14
- d. 22
- e. 8

32. Questions

215, 250, 320, 425, 565, ?

- a. 560
- b. 425
- c. 740
- d. 625
- e. 900

33. Questions

50, ?, 122, 170, 226

- a. 82
- b. 75
- c. 80
- d. 90
- e. 56

34. Questions

1, 2, ?, 16, 65, 326

- a. 3
- b. 8
- c. 12
- d. 5
- e. 10

35. Questions

120, ?, 60, 90, 180, 450

- a. 72
- b. 55
- c. 65
- d. 80
- e. 60

36. Questions

Find out the wrong number in the following number series.

314, 325, 306, 330, 298, 338

- a. 306
- b. 325
- c. 330
- d. 298
- e. 338

37. Questions

145, 141, 134, 116, 91, 55

- a. 134
- b. 116
- c. 91
- d. 145
- e. 55

38. Questions

3, 6, 18, 30, 102, 462

- a. 30
- b. 102
- c. 462
- d. 6
- e. 18

39. Questions

10, 32, 66, 127, 218, 345

- a. 10
- b. 66
- c. 127
- d. 32
- e. 345

40. Questions

240, 2400, 300, 1800, 455, 900

- a. 900
- b. 300
- c. 1800
- d. 455
- e. 240

41. Questions

The following question contains two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

i). $x^2 - 26x + 168 = 0$

ii). $y^2 - 21y + 108 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

42. Questions

i). $x^2 - 45x + 500 = 0$

ii). $y^2 - 35y + 306 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

43. Questions

i). $x^2 + 11x + 28 = 0$

ii). $y^2 - 8y + 16 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

44. Questions

i). $x^2 - 5x - 644 = 0$

ii). $y^2 - 17y + 66 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

45. Questions

i). $x^2 + 15x + 36 = 0$

ii). $y^2 + 5y + 6 = 0$

- a. $x > y$

- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

Explanations:

1. Questions

In hospital A:

Total number of patients admitted = $6000 * 30/100 = 1800$

Number of male patients admitted = 900

Number of female patients admitted = $(1800 - 900) = 900$

In hospital B:

Total number of patients admitted = $6000 * 24/100 = 1440$

Number of male patients admitted = 740

Number of female patients admitted = $(1440 - 740) = 700$

In hospital C:

Total number of patients admitted = $6000 * 16/100 = 960$

Number of male patients admitted = 500

Number of female patients admitted = $(960 - 500) = 460$

In hospital D:

Total number of patients admitted = $6000 * 10/100 = 600$

Number of male patients admitted = 200

Number of female patients admitted = $(600 - 200) = 400$

In hospital E:

Total number of patients admitted = $6000 * 20/100 = 1200$

Number of male patients admitted = 550

Number of female patients admitted = $(1200 - 550) = 650$

Hospital	Total number of patients admitted	Number of male patients admitted	Number of female patients admitted
A	1800	900	900
B	1440	740	700
C	960	500	460
D	600	200	400
E	1200	550	650

Answer: C

The total number of critical patients admitted in hospital A = $1800 \times \frac{20}{100} = 360$

Hence, the correct answer is option C

2. Questions

In hospital A:

Total number of patients admitted = $6000 \times \frac{30}{100} = 1800$

Number of male patients admitted = 900

Number of female patients admitted = $(1800 - 900) = 900$

In hospital B:

Total number of patients admitted = $6000 \times \frac{24}{100} = 1440$

Number of male patients admitted = 740

Number of female patients admitted = $(1440 - 740) = 700$

In hospital C:

Total number of patients admitted = $6000 \times \frac{16}{100} = 960$

Number of male patients admitted = 500

Number of female patients admitted = $(960 - 500) = 460$

In hospital D:

Total number of patients admitted = $6000 \times \frac{10}{100} = 600$

Number of male patients admitted = 200

Number of female patients admitted = $(600 - 200) = 400$

In hospital E:

Total number of patients admitted = $6000 \times \frac{20}{100} = 1200$

Number of male patients admitted = 550

Number of female patients admitted = $(1200 - 550) = 650$

Hospital	Total number of patients admitted	Number of male patients admitted	Number of female patients admitted
A	1800	900	900
B	1440	740	700
C	960	500	460
D	600	200	400
E	1200	550	650

Answer: D

The total number of patients admitted in hospital E = 1200

The number of female patients admitted in hospital D = 400

Required ratio = $1200 : 400 = 3 : 1$

Hence, the correct answer is option D

3. Questions

In hospital A:

Total number of patients admitted = $6000 \times 30/100 = 1800$

Number of male patients admitted = 900

Number of female patients admitted = $(1800 - 900) = 900$

In hospital B:

Total number of patients admitted = $6000 \times 24/100 = 1440$

Number of male patients admitted = 740

Number of female patients admitted = $(1440 - 740) = 700$

In hospital C:

Total number of patients admitted = $6000 \times 16/100 = 960$

Number of male patients admitted = 500

Number of female patients admitted = $(960 - 500) = 460$

In hospital D:

Total number of patients admitted = $6000 \times 10/100 = 600$

Number of male patients admitted = 200

Number of female patients admitted = $(600 - 200) = 400$

In hospital E:

Total number of patients admitted = $6000 * 20/100 = 1200$

Number of male patients admitted = 550

Number of female patients admitted = $(1200 - 550) = 650$

Hospital	Total number of patients admitted	Number of male patients admitted	Number of female patients admitted
A	1800	900	900
B	1440	740	700
C	960	500	460
D	600	200	400
E	1200	550	650

Answer: E

The total number of patients admitted in hospital F = $(1800 + 72) = 1872$

$1872 = 1440 * (100 + m + 20)/100$

$130 = (120 + m)$

$m = 10$

Required answer = $2m = 2 * 10 = 20$

Hence, the correct answer is option E

4. Questions

In hospital A:

Total number of patients admitted = $6000 * 30/100 = 1800$

Number of male patients admitted = 900

Number of female patients admitted = $(1800 - 900) = 900$

In hospital B:

Total number of patients admitted = $6000 * 24/100 = 1440$

Number of male patients admitted = 740

Number of female patients admitted = $(1440 - 740) = 700$

In hospital C:

Total number of patients admitted = $6000 * 16/100 = 960$

Number of male patients admitted = 500

Number of female patients admitted = $(960 - 500) = 460$

In hospital D:

Total number of patients admitted = $6000 * 10/100 = 600$

Number of male patients admitted = 200

Number of female patients admitted = $(600 - 200) = 400$

In hospital E:

Total number of patients admitted = $6000 * 20/100 = 1200$

Number of male patients admitted = 550

Number of female patients admitted = $(1200 - 550) = 650$

Hospital	Total number of patients admitted	Number of male patients admitted	Number of female patients admitted
A	1800	900	900
B	1440	740	700
C	960	500	460
D	600	200	400
E	1200	550	650

Answer: E

The number of male patients admitted in hospital C = 500

The number of male patients admitted for cancer in hospital C = $500 * 3/5 = 300$

Hence, the correct answer is option E

5. Questions**In hospital A:**

Total number of patients admitted = $6000 * 30/100 = 1800$

Number of male patients admitted = 900

Number of female patients admitted = $(1800 - 900) = 900$

In hospital B:

Total number of patients admitted = $6000 * 24/100 = 1440$

Number of male patients admitted = 740

Number of female patients admitted = $(1440 - 740) = 700$

In hospital C:

Total number of patients admitted = $6000 * 16/100 = 960$

Number of male patients admitted = 500

Number of female patients admitted = $(960 - 500) = 460$

In hospital D:

Total number of patients admitted = $6000 * 10/100 = 600$

Number of male patients admitted = 200

Number of female patients admitted = $(600 - 200) = 400$

In hospital E:

Total number of patients admitted = $6000 * 20/100 = 1200$

Number of male patients admitted = 550

Number of female patients admitted = $(1200 - 550) = 650$

Hospital	Total number of patients admitted	Number of male patients admitted	Number of female patients admitted
A	1800	900	900
B	1440	740	700
C	960	500	460
D	600	200	400
E	1200	550	650

Answer: D

The sum of the number of female patients who are admitted in hospitals A, B and D together = $(900 + 700 + 400) = 2000$

Hence, the correct answer is option D

6. Questions

In class P:

The total number of students who prefer boating and game zone = $350 * 2 = 700$

The number of students who prefer boating = 250

The number of students who prefer game zone = $(700 - 250) = 450$

In class Q:

The total number of students who prefer boating and game zone = $400 * 2 = 800$

The number of students who prefer boating = 300

The number of students who prefer game zone = $(800 - 300) = 500$

In class R:

The total number of students who prefer boating and game zone = $500 * 2 = 1000$

The number of students who prefer boating = 350

The number of students who prefer game zone = $(1000 - 350) = 650$

In class S:

The total number of students who prefer boating and game zone = $200 * 2 = 400$

The number of students who prefer boating = 100

The number of students who prefer game zone = $(400 - 100) = 300$

Class	The total number of students who prefer boating and game zone	The number of students who prefer boating	The number of students who prefer game zone
P	700	250	450
Q	800	300	500
R	1000	350	650
S	400	100	300

Answer: D

Let, the number of male students who prefer boating and game zone in class Q be $2x$ and x respectively.

Let, the number of female students who prefer boating and game zone in class Q be y and $4y$ respectively

$$2x + y = 300 \text{ ----> (1)}$$

$$x + 4y = 500 \text{ -----> (2)}$$

By solving equation (1) and (2), we get

$$x = 100 \text{ and } y = 100$$

The number of female students who prefer game zone in class Q = $4 * 100 = 400$

Hence, the correct answer is option D

7. Questions

In class P:

The total number of students who prefer boating and game zone = $350 * 2 = 700$

The number of students who prefer boating = 250

The number of students who prefer game zone = $(700 - 250) = 450$

In class Q:

The total number of students who prefer boating and game zone = $400 * 2 = 800$

The number of students who prefer boating = 300

The number of students who prefer game zone = $(800 - 300) = 500$

In class R:

The total number of students who prefer boating and game zone = $500 * 2 = 1000$

The number of students who prefer boating = 350

The number of students who prefer game zone = $(1000 - 350) = 650$

In class S:

The total number of students who prefer boating and game zone = $200 * 2 = 400$

The number of students who prefer boating = 100

The number of students who prefer game zone = $(400 - 100) = 300$

Class	The total number of students who prefer boating and game zone	The number of students who prefer boating	The number of students who prefer game zone
P	700	250	450
Q	800	300	500
R	1000	350	650
S	400	100	300

Answer: C

The number of students who prefer boating in class Q = 300

The number of students who prefer game zone in class S = 300

Required percentage = $(300/300) * 100 = 100\%$

Hence, the correct answer is option C

8. Questions

In class P:

The total number of students who prefer boating and game zone = $350 * 2 = 700$

The number of students who prefer boating = 250

The number of students who prefer game zone = $(700 - 250) = 450$

In class Q:

The total number of students who prefer boating and game zone = $400 * 2 = 800$

The number of students who prefer boating = 300

The number of students who prefer game zone = $(800 - 300) = 500$

In class R:

The total number of students who prefer boating and game zone = $500 * 2 = 1000$

The number of students who prefer boating = 350

The number of students who prefer game zone = $(1000 - 350) = 650$

In class S:

The total number of students who prefer boating and game zone = $200 * 2 = 400$

The number of students who prefer boating = 100

The number of students who prefer game zone = $(400 - 100) = 300$

Class	The total number of students who prefer boating and game zone	The number of students who prefer boating	The number of students who prefer game zone
P	700	250	450
Q	800	300	500
R	1000	350	650
S	400	100	300

Answer: B

The number of students who prefer boating in class T = $(350 + 50) = 400$

The number of students who prefer game zone in class T = $500 * 3/5 = 300$

Required sum = $(400 + 300) = 700$

Hence, the correct answer is option B

9. Questions

In class P:

The total number of students who prefer boating and game zone = $350 * 2 = 700$

The number of students who prefer boating = 250

The number of students who prefer game zone = $(700 - 250) = 450$

In class Q:

The total number of students who prefer boating and game zone = $400 * 2 = 800$

The number of students who prefer boating = 300

The number of students who prefer game zone = $(800 - 300) = 500$

In class R:

The total number of students who prefer boating and game zone = $500 * 2 = 1000$

The number of students who prefer boating = 350

The number of students who prefer game zone = $(1000 - 350) = 650$

In class S:

The total number of students who prefer boating and game zone = $200 * 2 = 400$

The number of students who prefer boating = 100

The number of students who prefer game zone = $(400 - 100) = 300$

Class	The total number of students who prefer boating and game zone	The number of students who prefer boating	The number of students who prefer game zone
P	700	250	450
Q	800	300	500
R	1000	350	650
S	400	100	300

Answer: D

The total number of students prefer boating and game zone in P = 700

The number of students prefer game zone in R = 650

Required difference = $(700 - 650) = 50$

Hence, the correct answer is option D

10. Questions

In class P:

The total number of students who prefer boating and game zone = $350 * 2 = 700$

The number of students who prefer boating = 250

The number of students who prefer game zone = $(700 - 250) = 450$

In class Q:

The total number of students who prefer boating and game zone = $400 * 2 = 800$

The number of students who prefer boating = 300

The number of students who prefer game zone = $(800 - 300) = 500$

In class R:

The total number of students who prefer boating and game zone = $500 * 2 = 1000$

The number of students who prefer boating = 350

The number of students who prefer game zone = $(1000 - 350) = 650$

In class S:

The total number of students who prefer boating and game zone = $200 * 2 = 400$

The number of students who prefer boating = 100

The number of students who prefer game zone = $(400 - 100) = 300$

Class	The total number of students who prefer boating and game zone	The number of students who prefer boating	The number of students who prefer game zone
P	700	250	450
Q	800	300	500
R	1000	350	650
S	400	100	300

Answer: A

$$(5m + 50) = 450$$

$$5m = 400$$

$$m = 80$$

Hence, the correct answer is option A

11. Questions

Answer: A

Let, the speed of the train = x m/s

The length of the train = y m

$$y = (x + 5) * 3 \text{ ----> (1)}$$

$$y = (x - 15) * 5 \text{ ----> (2)}$$

$$(x + 5) * 3 = (x - 15) * 5$$

$$3x + 15 = 5x - 75$$

$$2x = 90$$

$$x = 45$$

The speed of the train = 45 m/s

Hence, the correct answer is option A

12. Questions

Answer: D

The total number of students in school A = $[(100 + x)/100] * \text{The total number of students in school B}$.

The total number of students in school B = $780 * [10/13] = 600$

$$780 = [(100 + x)/100] * 600$$

$$130 = (100 + x)$$

$$x = 30$$

Hence, the correct answer is option D

13. Questions

Answer: B

Let, the present ages of Mano and Mohan are $3x$ years and $2x$ years respectively.

$$\text{Mano} + \text{Mohan} + \text{Madhan} = 21 * 3 = 63 \text{ years}$$

$$3x + 2x + 3x + 7 = 63$$

$$8x = 56$$

$$x = 7$$

The present age of Mohan = $2 * 7 = 14$ years

Hence, the correct answer is option B

14. Questions

Answer: A

$$SI = PNR/100$$

$$1200 = 4000 * R * 2/100$$

$$R = 15\%$$

Therefore, according to the question, for CI the rate of interest is $R+5 = 20\%$.

$$CI = P * (1 + R/100)^n - P$$

$$CI = 8800 * 1.2 * 1.2 - 8800$$

$$CI = 12672 - 8800$$

$$CI = \text{Rs. } 3872$$

Hence, the correct answer is option A

15. Questions

Answer: A

Let, the cost price of the pants = Rs. x

The cost price of the shirts = Rs. $(4000 - x)$

$$x * 120/100 + (4000 - x) * 90/100 = 4350$$

$$1.2x + 3600 - 0.9x = 4350$$

$$0.3x + 3600 = 4350$$

$$0.3x = 750$$

$$x = 2500$$

The cost price of the shirts = $(4000 - 2500) = \text{Rs. } 1500$

Hence, the correct answer is option A

16. Questions

Answer: D

The time taken by A alone to complete the work = $10 * 5 = 50$ hours

The time taken by B alone to complete the work = $5 * 5 = 25$ hours

Let, the total work = 50 units (LCM of 50 and 25)

The efficiency of A = $50/50 = 1$ unit/hour

The efficiency of B = $50/25 = 2$ units/hour

The efficiency of C = $2 * 2 = 4$ units/hour

Required time taken = $[50/(1 + 4)] * 2$

$$= 50/5 * 2 = 20 \text{ hours}$$

Hence, the correct answer is option D

17. Questions

Answer: C

The ratio of the profit share of Amit to Balu = $[(5x * 5) + (5x + 3000) * 7] : (4x * 12)$

$$= (25x + 35x + 21000) : (48x)$$

$$= (60x + 21000) : 48x$$

$$= (20x + 7000) : 16x$$

$$= (10x + 3500) : 8x$$

$$= (5x + 1750) : 4x$$

$$(5x + 1750)/4x = 55/16$$

$$(x + 350)/x = 11/4$$

$$4x + 1400 = 11x$$

$$7x = 1400$$

$$x = 200$$

The initial investment of Amit = $5 * 200 = \text{Rs. } 1000$

Hence, the correct answer is option C

18. Questions

Answer: C

The area of the circle = $\pi r^2 \text{ cm}^2$

$$616 = (22/7) * r^2$$

$$28 * 7 = r^2$$

$$r^2 = 196$$

$$r = 14 \text{ cm}$$

The length of the rectangle = $(14 + 6) = 20 \text{ cm}$

The perimeter of the rectangle = $2(l + b) \text{ cm}$

$$74 = 2(20 + b)$$

$$b = 17 \text{ cm}$$

Hence, the correct answer is option C

19. Questions

Answer: D

Let, the number A and C be $7x$ and $3x$ respectively.

$$B = 340/100 * 3x$$

$$B = 51x/5$$

The ratio of the value of A, B to C = $35: 51: 15$

$$35x + 15x = 51x - 40$$

$$50x = 51x - 40$$

$$x = 40$$

$$\text{The value of C} = 15 * 40 = 600$$

Hence, the correct answer is option D

20. Questions

Answer: A

The quantity of water in mixture A = 200 litres

The quantity of milk in mixture A = $200 * 5/4 = 250$ litres

The quantity of milk in mixture B = $(250 - 100) = 150$ litres

The quantity of water in mixture B = $150 * 4/3 = 200$ litres

Required sum = $(200 + 150) = 350$ litres

Hence, the correct answer is option A

21. Questions**Answer: A**

$$25\% \text{ of } 720 + \sqrt{625} - 40\% \text{ of } 1200 = ?$$

$$180 + 25 - 480 = ?$$

$$? = -275$$

Hence, the correct answer is option A

22. Questions**Answer: D**

$$(\sqrt{576} + 276) / 5 + 30\% \text{ of } 800 = ? * 2$$

$$(24 + 276) / 5 + 240 = ? * 2$$

$$300 / 5 + 240 = ? * 2$$

$$60 + 240 = ? * 2$$

$$300 / 2 = ?$$

$$? = 150$$

Hence, the correct answer is option D

23. Questions**Answer: D**

$$2400 * 5 / 16 + 54 * 3 = ?^2 + 12$$

$$150 * 5 + 162 = ?^2 + 12$$

$$750 + 162 = ?^2 + 12$$

$$912 - 12 = ?^2$$

$$?^2 = 900$$

$$? = 30$$

Hence, the correct answer is option D

24. Questions**Answer: A**

$$60\% \text{ of } 200 - 10\% \text{ of } 500 + 35 = ? * 5$$

$$120 - 50 + 35 = ? * 5$$

$$70 + 35 = ? * 5$$

$$105 / 5 = ?$$

$$? = 21$$

Hence, the correct answer is option A

25. Questions

Answer: C

$$156 + 244 - 18 * 7 + ? = 80 * 5$$

$$400 - 126 + ? = 400$$

$$274 + ? = 400$$

$$? = 126$$

Hence, the correct answer is option C

26. Questions

Answer: D

$$(1332)^{1/3} * 3.98 + ? = 32.12\% \text{ of } 299.99$$

$$(1331)^{1/3} * 4 + ? = 32\% \text{ of } 300$$

$$11 * 4 + ? = 96$$

$$44 + ? = 96$$

$$? = 52$$

Hence, the correct answer is option D

27. Questions

Answer: C

$$[36.21 * 5/6] + 44.89 + 21.2 * 1.97 = ?$$

$$[36 * 5/6] + 45 + 21 * 2 = ?$$

$$6 * 5 + 45 + 42 = ?$$

$$30 + 45 + 42 = ?$$

$$? = 117$$

Hence, the correct answer is option C

28. Questions

Answer: C

$$29.99\% \text{ of } 400.19 + 5.12^2 = ?^3 - 71.3$$

$$30\% \text{ of } 400 + 5^2 = ?^3 - 71$$

$$120 + 25 = ?^3 - 71$$

$$145 + 71 = ?^3$$

$$?^3 = 216$$

$$? = 6$$

Hence, the correct answer is option C

29. Questions

Answer: E

$$\sqrt[3]{64} \cdot 23 * 12.13 + 179.98 - ? = 14.12 * 9.89$$

$$\sqrt[3]{64} * 12 + 180 - ? = 14 * 10$$

$$4 * 12 + 180 - ? = 140$$

$$48 + 180 - ? = 140$$

$$228 - ? = 140$$

$$? = 88$$

Hence, the correct answer is option E

30. Questions

Answer: B

$$15.12 * (25)^{1/2} - 18.01\% \text{ of } 200.12 = ?$$

$$15 * 5 - 18\% \text{ of } 200 = ?$$

$$75 - 36 = ?$$

$$? = 39$$

Hence, the correct answer is option B

31. Questions

Answer: E

$$8 * 7 = 56$$

$$56/2 = 28$$

$$28 * 7 = 196$$

$$196/2 = 98$$

$$98 * 7 = 686$$

Hence, the correct answer is option E

32. Questions

Answer: C

$$215 + 35 = 250$$

$$250 + 70 = 320$$

$$320 + 105 = 425$$

$$425 + 140 = 565$$

$$565 + 175 = \mathbf{740}$$

Hence, the correct answer is option C

33. Questions

Answer: A

$$7^2 + 1 = 50$$

$$9^2 + 1 = \mathbf{82}$$

$$11^2 + 1 = 122$$

$$13^2 + 1 = 170$$

$$15^2 + 1 = 226$$

Hence, the correct answer is option A

34. Questions

Answer: D

$$(1 * 1) + 1 = 2$$

$$(2 * 2) + 1 = \mathbf{5}$$

$$(5 * 3) + 1 = 16$$

$$(16 * 4) + 1 = 65$$

$$(65 * 5) + 1 = 326$$

Hence, the correct answer is option D

35. Questions

Answer: E

$$120 * 0.5 = \mathbf{60}$$

$$60 * 1 = 60$$

$$60 * 1.5 = 90$$

$$90 * 2 = 180$$

$$180 * 2.5 = 450$$

Hence, the correct answer is option E

36. Questions**Answer: B**

$$314 + 8 = 322$$

$$322 - 16 = 306$$

$$306 + 24 = 330$$

$$330 - 32 = 298$$

$$298 + 40 = 338$$

Hence, the correct answer is option B

37. Questions**Answer: A**

$$145 - 2^2 = 141$$

$$141 - 3^2 = 132$$

$$132 - 4^2 = 116$$

$$116 - 5^2 = 91$$

$$91 - 6^2 = 55$$

Hence, the correct answer is option A

38. Questions**Answer: E**

$$3 \quad 6 \quad 12 \quad 30 \quad 102 \quad 462$$

$$3 \quad 6 \quad 18 \quad 72 \quad 360$$

$$*2 \quad *3 \quad *4 \quad *5$$

Hence, the correct answer is option E

39. Questions**Answer: D**

$$2^3 + 2 = 10$$

$$3^3 + 2 = 29$$

$$4^3 + 2 = 66$$

$$5^3 + 2 = 127$$

$$6^3 + 2 = 218$$

$$7^3 + 2 = 345$$

Hence, the correct answer is option D

40. Questions

Answer: D

$$240 * 10 = 2400$$

$$2400/8 = 300$$

$$300 * 6 = 1800$$

$$1800/4 = \mathbf{450}$$

$$450 * 2 = 900$$

Hence, the correct answer is option D

41. Questions

Answer: B

$$\text{i). } x^2 - 26x + 168 = 0$$

$$x^2 - 12x - 14x + 168 = 0$$

$$x(x - 12) - 14(x - 12) = 0$$

$$(x - 12)(x - 14) = 0$$

$$x = 12, 14$$

$$\text{ii). } y^2 - 21y + 108 = 0$$

$$y^2 - 12y - 9y + 108 = 0$$

$$y(y - 12) - 9(y - 12) = 0$$

$$(y - 12)(y - 9) = 0$$

$$y = 12, 9$$

Thus, $x \geq y$

Hence, the correct answer is option B

42. Questions

Answer: A

$$\text{i). } x^2 - 45x + 500 = 0$$

$$x^2 - 20x - 25x + 500 = 0$$

$$x(x - 20) - 25(x - 20) = 0$$

$$(x - 20)(x - 25) = 0$$

$$x = 20, 25$$

$$\text{ii). } y^2 - 35y + 306 = 0$$

$$y^2 - 18y - 17y + 306 = 0$$

$$y(y - 18) - 17(y - 18) = 0$$

$$(y - 18)(y - 17) = 0$$

$$y = 18, 17$$

Thus, $x > y$

Hence, the correct answer is option A

43. Questions

Answer: D

$$\text{i). } x^2 + 11x + 28 = 0$$

$$x^2 + 7x + 4x + 28 = 0$$

$$x(x + 7) + 4(x + 7) = 0$$

$$(x + 7)(x + 4) = 0$$

$$x = -7, -4$$

$$\text{ii). } y^2 - 8y + 16 = 0$$

$$y^2 - 4y - 4y + 16 = 0$$

$$y(y - 4) - 4(y - 4) = 0$$

$$(y - 4)(y - 4) = 0$$

$$y = 4, 4$$

Thus, $x < y$

Hence, the correct answer is option D

44. Questions

Answer: C

$$\text{i). } x^2 - 5x - 644 = 0$$

$$x^2 - 28x + 23x - 644 = 0$$

$$x(x - 28) + 23(x - 28) = 0$$

$$(x - 28)(x + 23) = 0$$

$$x = 28, -23$$

$$\text{ii). } y^2 - 17y + 66 = 0$$

$$Y^2 - 11y - 6y + 66 = 0$$

$$y(y - 11) - 6(y - 11) = 0$$

$$(y - 11)(y - 6) = 0$$

$$y = 11, 6$$

Thus, the relationship between x and y cannot be determined.

Hence, the correct answer is option C

45. Questions

Answer: E

$$\text{i). } x^2 + 15x + 36 = 0$$

$$x^2 + 12x + 3x + 36 = 0$$

$$x(x + 12) + 3(x + 12) = 0$$

$$(x + 12)(x + 3) = 0$$

$$x = -12, -3$$

$$\text{ii). } y^2 + 5y + 6 = 0$$

$$y^2 + 2y + 3y + 6 = 0$$

$$y(y + 2) + 3(y + 2) = 0$$

$$(y + 2)(y + 3) = 0$$

$$y = -2, -3$$

Thus, $x \leq y$

Hence, the correct answer is option E